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Linux and shell programming

Lecture 1: Introduction to Linux

© spring 2025 – **Dr. Dina Awny**

References

- William E. Jr. Shotts, The Linux Command Line, 2nd Edition: A Complete Introduction Paperback – Illustrated, 7 March 2019
  - Linux Command Line Books by William Shotts
- Javapoints
  - Linux Shell - javatpoint
- Tutorialspoint
  - Linux/Unix Tutorial
  - Introduction to Linux Shell and Shell Scripting - GeeksforGeeks
- W3school
  - https://w3schools.tech/tutorial/unix/index
- Download Ubuntu
  - Download Ubuntu Desktop | Ubuntu
- Download VirtualBox
  - Downloads – Oracle VirtualBox

LMS

- Course Materials: Lectures and Lab Assignments
- Online Quizzes
- Project Submission
- Lab Assignment Submission

Grades

- Midterm: 15 Degree
- Activities
  - Project: 20 Degree
  - Quizzes (Practical/LMS): 20 Degree (6 quizzes, Inside lab, best 5)
  - Practical Assignments: 20 Degree (LMS + Discussion)
- Final Exam: 25 Degree

Calendar

- Week 1: Introduction to Linux
- Week 2: Linux Directories and Files
- Week 3: Getting help and filters
- Week 4: System accounts and groups
- Week 5: Linux Process and runlevels
- Week 6: Basics for Shell Programming
- Week 7: Midterm

Calendar

- Week 8: Loops and Functions
- Week 9: Shell Functions and arrays
- Week 10: I/O Redirection
- Week 11: File links and Archive
- Week 12: Revision and Case Studies
- Week 13: Revision and Case Studies + Project Discussion
- Week 14: Practical Discussion

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- Linux Kernel and shell
- Linux VS Unix
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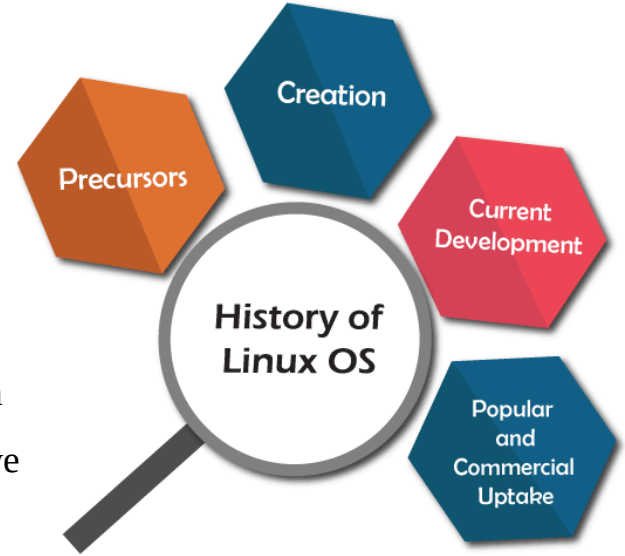
Introduction to Linux

- Linux is an open-source operating system-based family on the Linux kernel.
- Open Source: Anyone can view, modify, and distribute the source code
- Multi-user: Supports multiple users simultaneously
- Multitasking: Can run multiple programs at the same time
- Security: Highly secure due to its architecture and active community
- Stability: Known for its robust and crash-resistant nature
- Flexibility: Can be customized for various purposes

History of Linux OS



•Precursors:

- Unix was created in 1969 at AT&T's Bell Labs.
- Initially written in assembly language, Unix was later rewritten in C by Dennis Ritchie in 1973, making it easier to port to different computer platforms.

•Creation of Linux:

- Linus Torvalds, inspired by a Unix course and MINIX operating system, began developing the Linux kernel in 1991 due to frustrations with MINIX's restrictive licensing.
- The Linux kernel initially relied on MINIX software but transitioned to GNU components due to the GNU Project's free code and the GNU General Public License (GPL).

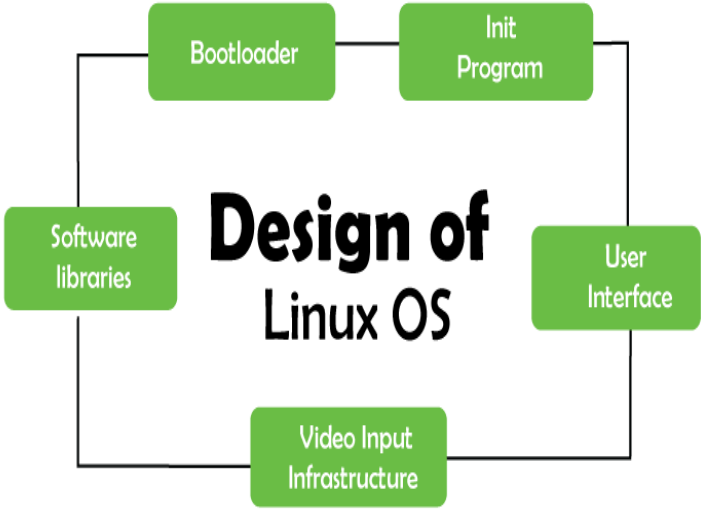
History of Linux OS cont.

- **Current Development:**
  - The Linux Kernel is maintained by Greg Kroah-Hartman, with GNU components supported by the Free Software Foundation led by William John Sullivan.
  - Linux distributions combine the Linux kernel, GNU components, third-party software, and package management tools, developed by the Linux community, corporations, and individuals.
- **Popular and Commercial Adoption:**
  - Linux gained adoption in the mid-1990s, particularly in supercomputing and later in commercial environments with IBM, Dell, and HP supporting Linux to challenge Microsoft.
  - Linux is widely used in embedded systems, supercomputers, servers, enterprise systems, and home desktops.

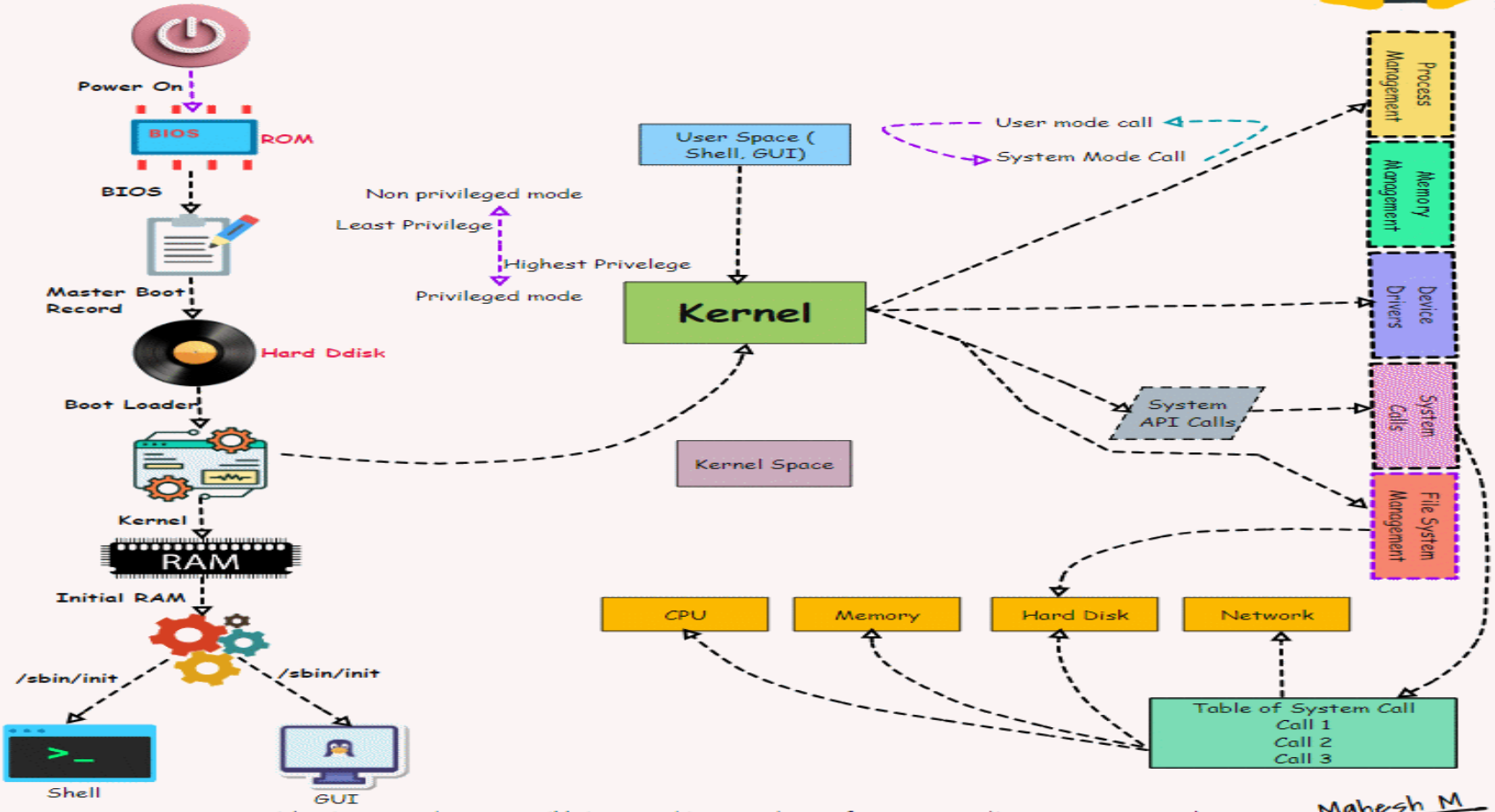
Design of Linux OS

- A ***bootloader***, for instance, systemd-boot, SYSLINUX, LILO, and GNU GRUB. It is a program that can load the Linux kernel into the main memory.
- An ***init program,*** like the traditional sysvinit and the newer Upstart, OpenRC, and systemd. It is the first process announced by the Linux kernel and the process tree root.
- ***Software libraries***, which include code that can be applied by running processes.
- **User interface**

  Also, the user interface is called a shell. It is either a GUI (graphical user interface), a CLI (command-line interface), or controls attached to the related hardware, which is normal for embedded systems.



HOW LINUX WORKS



/sbin/init now directs to /lib/systemd/systemd, signifying systemd's initiation control.

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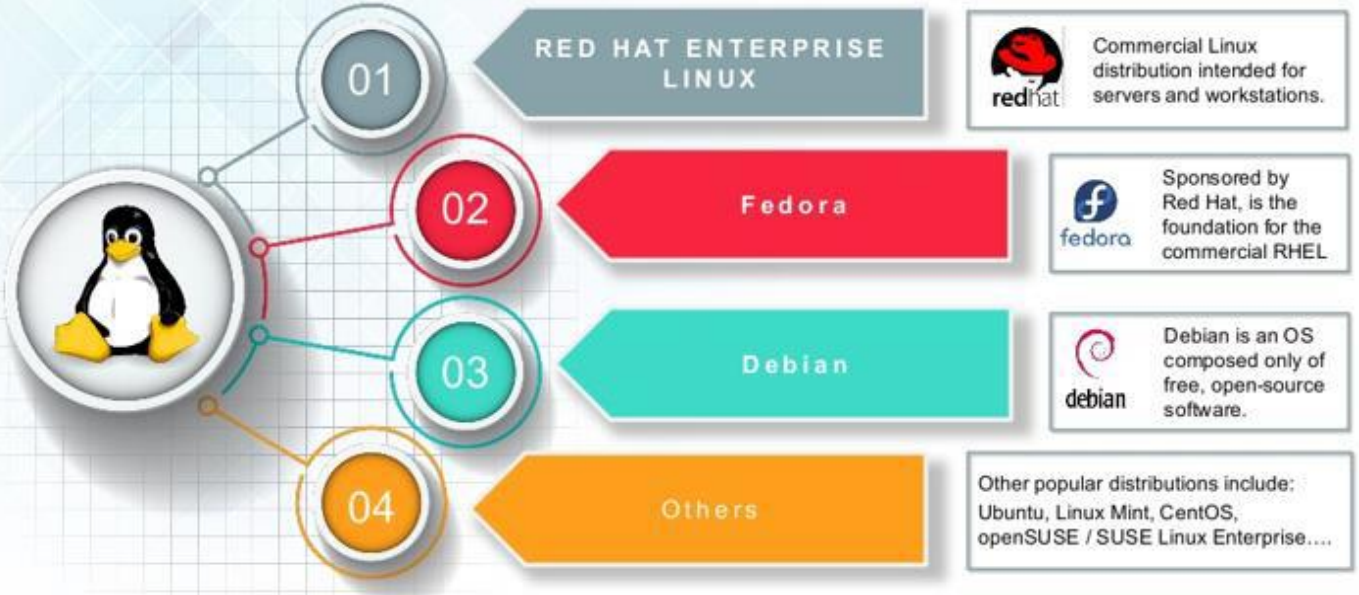
Linux distributions

Various Linux Distributions



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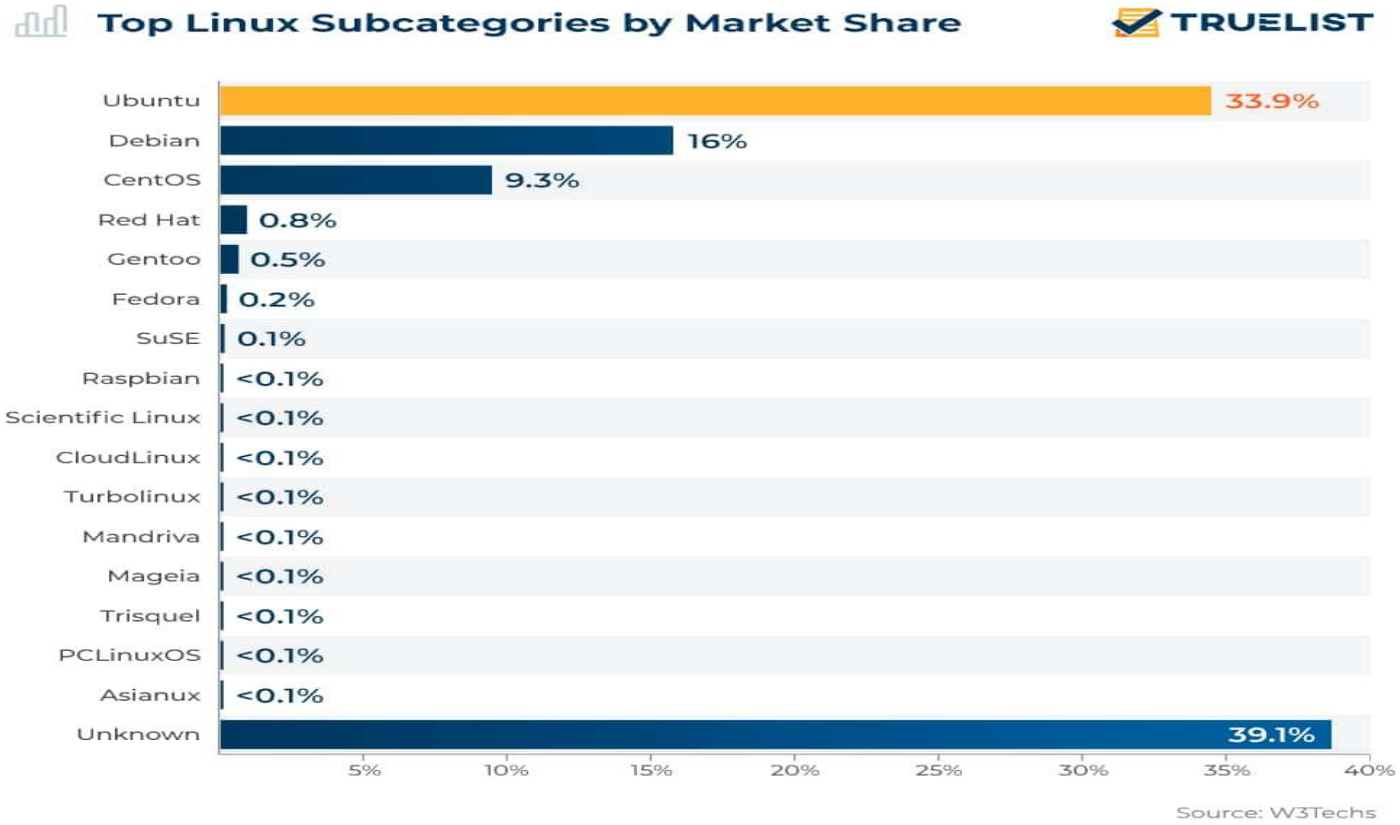
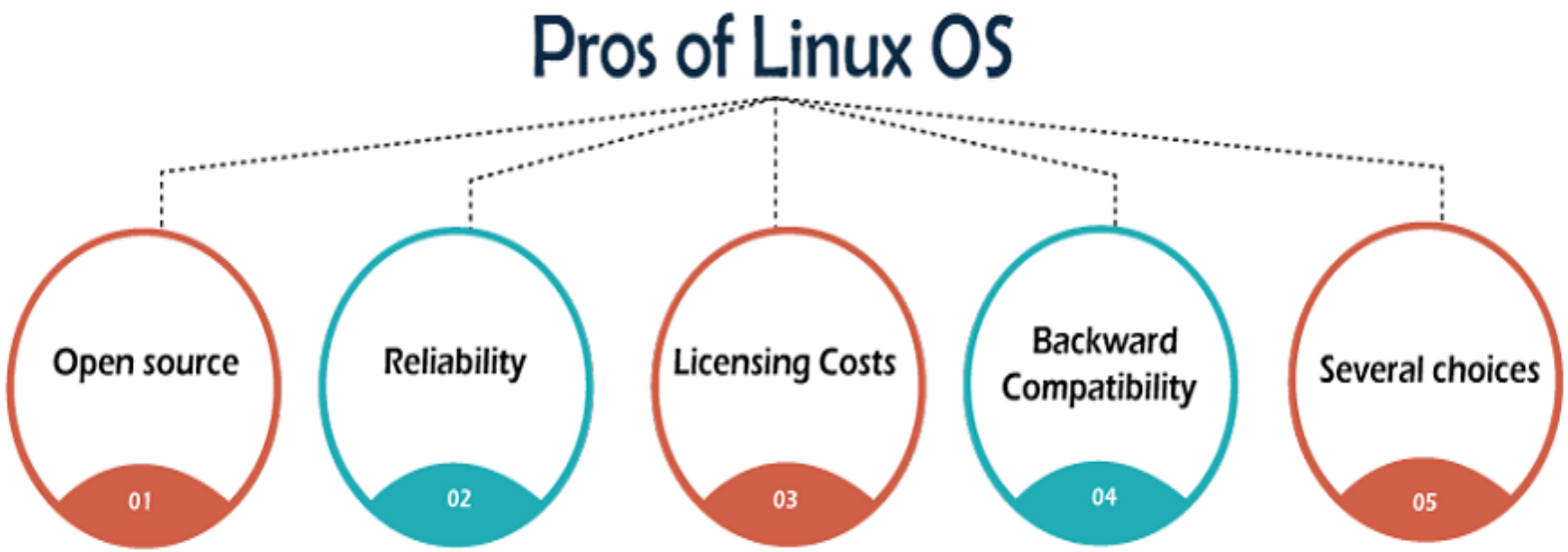
Top Linux distributions



Pros of Linux OS



Cons of Linux OS

Cons of Linux OS

- Lack of Standard
- Support Costs
- Proprietary Software
- Steep Learning Curve
- Unsupported Hardware

Linux kernel and shell

- ***Kernel*** is the heart of the Linux operating system. It manages the resources of Linux such as:
  - File management
  - Multitasking
  - Memory and I/O management
  - Process management
  - Device management
  - Networking support

Linux shell

- The shell is a program that allows the user to type commands, options, and arguments
- Many shell programs exist
- It is a command language interpreter that executes commands read from the standard input device such as keyboard or from a file.
- Most popular shell is the “Bash” (Bourne Again Shell)
- A user’s shell can be changed by the usermod command by root
- User’s default shell stored in /etc/passwd

In Unix, there are two major types of shells −

- Bourne shell − If you are using a Bourne-type shell, the $ character is the default prompt.
- C shell − If you are using a C-type shell, the % character is the default prompt.

Linux shell cont.

The Bourne Shell has the following subcategories −

- Bourne shell (sh) : Bourne shell was the first shell to appear on Unix systems.
- It is usually installed as **/bin/sh** on most versions of Unix.
- Korn shell (ksh)
- Bourne Again shell (bash) : an enhanced version of the original Unix shell program (sh).
- POSIX shell (sh)

The different C-type shells follow −

- C shell (csh): a shell with a syntax close to the C language syntax.
- TENEX/TOPS C shell (tcsh)

➢ To find all of the available shells on your system, print the contents of

the file “/etc/shells” using the “cat” command

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The bash Shell

- The Bash is a **command language interpreter** as well as a **programming language**.
- It supports **variables, functions, and flow control**, like other programming languages.
- It can also read and execute the commands from a file, which is called a **shell script.**
- Bash offers various functional improvements over Bourne Shell (sh) for both interactive and programming use.
- The Bash contains the following improvements over sh:
  - It provides command-line editing
  - It contains unlimited size command history
  - It provides Job Control
  - It facilitates with Shell Functions and Aliases
  - It provides the indexed arrays of unlimited size
  - It contains integer arithmetic in any base from 2 to 64.

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Features of Bash

Some key features of Bash are as follows:

•**Shell Syntax:** The shell syntax contains **shell operations**, **quoting**, and **comments**.

•**Shell commands:** Shell commands are the types of commands that you can execute. These commands can be simple commands, pipelines, lists, compound commands, and more.

•**Shell Functions:** Shell functions are used to group commands by name.

•**Shell parameters:** Basically, a parameter is an entity that stores value; it can be a name, number, or special character.

•**Shell Expansions:** Shell expansion is a technique that is used by Bash to expand the parameters. Expansion is performed on the command line after the input has been splitted into tokens.

•**Redirections:** It is a way to manage and control the input and output.

•**Command execution:** It decides how the system will react when we execute a command.

•**Shell Scripts:** It is a text file that has shell commands and executes them when it is used

Shell commands

1. **An executable program** like all those files we saw in /usr/bin. Within this category, programs can be compiled binaries such as programs written in C and C++, or programs written in scripting languages such as the shell, python, etc.

2**. A command built into the shell itself** (built-in commands) bash supports a number of commands internally called shell built-ins.

3.**A shell function.** These are miniature shell scripts incorporated into the environment.

4. **An alias**. Commands that we can define ourselves, built from other commands.

Syntax: command [options…] [arguments…]

- Commands, options, and arguments are all case sensitive

Identify Shell commands

It is often useful to know exactly which of the four kinds of commands is
being used.

A. Display a Command's Type

- The ***type*** command is a shell built-in that displays the kind of command the shell will execute,

```
dina@dina-virtual-machine:~$ pwd
/home/dina
dina@dina-virtual-machine:~$ type pwd
pwd is a shell builtin
dina@dina-virtual-machine:~$ which pwd
/usr/bin/pwd
dina@dina-virtual-machine:~$ _
```

B. Display an Executable's Location

- Sometimes there is more than one version of an executable program installed on a system.
- To determine the exact location of a given executable, the ***which*** command is used.

```
[root@master ~]# pwd
/root
[root@master ~]# type pwd
pwd is a shell builtin
[root@master ~]# which pwd
/usr/bin/pwd
[root@master ~]#
```

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Unix Vs Linux

Comparison	Linux	Unix
Definition	It is an open-source operating system which is *freely available to everyone*.	It is an operating system which *can be only used by its copyrighters*.
Examples	It has different distros like Ubuntu, Redhat, Fedora, etc	IBM AIX, HP-UX and Sun Solaris.
Users	Nowadays, Linux is in great demand. Anyone can use Linux whether a home user, developer or a student.	It was developed mainly for servers, workstations and mainframes.
Usage	Linux is used everywhere from servers, PC, smartphones, tablets to mainframes and supercomputers.	It is used in servers, workstations and PCs.
Interface	The default interface is BASH (Bourne Again SHell). But some distros have developed their own interfaces.	It originally used Bourne shell. But is also compatible with other GUIs.

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Unix Vs Linux

File system support	Linux supports more file system than Unix.	It also supports file system but lesser than Linux.
Coding	Linux is a Unix clone,behaves like Unix but doesn't contain its code.	Unix contain a completely different coding developed by AT&T Labs.
Operating system	Linux is just the kernel.	Unix is a complete package of Operating system.
Security	It provides higher security. Linux has about 60-100 viruses listed till date.	Unix is also highly secured. It has about 85-120 viruses listed till date
Error detection and solution	As Linux is open-source,whenever a user post any kind of threat, developers from all over the world start working on it. And hence, it provides faster solution.	In Unix, users have to wait for some time for the problem to be resolved.

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Linux Vs Windows

Parameter	Linux	Windows
Access	Users can access the source code of kernel in Linux and can alter the kernel according to need.	Usually, users cannot access the source code. However, members of some groups can have access to it.
Variety	Linux has several distributions that are highly customizable.	Windows have fewer options to customize.
License	Linux is distributed under the **GPL(GNU General Public License) license**.	Windows is distributed under a **Proprietary commercial software license**.

Parameter	Linux	Windows
Command-line	The command line usually referred to **as Terminal**, which is the most useful tool of the Linux system. It used for administration and daily tasks. For the end-users, it does not look so effective.	Windows also have a command line, but it is not such effective as a comparison to the Linux terminal. Most users prefer the GUI options for daily tasks.
Written in	Linux is written in **assembly language** and **C**.	Windows is written in **C++** and **assembly language**.

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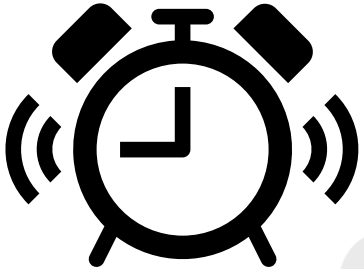
Linux Vs Windows

Parameter	Linux	Windows
Security	Linux OS is more secure than Windows. It is hard for the hackers and attackers to find a loophole in it. So, Linux is hard to breakthrough.	Windows is less secure than Linux. Attackers primarily target the Windows for malware and virus. Windows is most vulnerable without anti-virus.

Thank you !

Get ready for our Linux directories next week!